ACTION TRIGGERS · ENERGY, INFRASTRUCTURE & CLIMATE RESILIENCE · 17 JULY 2026

Grids Will Ration Power to Large Users, Starting With Data Centres

Grid operators have gained and rehearsed emergency powers to cut supply to data centres, so access to power is no longer a place in a queue but a condition that can be withdrawn. Water and grid equipment now limit new sites too.

Produced using **Shaping Tomorrow's six-stage process for converting external change into decision-ready intelligence**.
Scanned and filtered by AI. **Stress-tested and judged by humans**.

How long you have to get ready

The four actions below, from today to the moment each falls due. Select a pill to see what would change it, or a number to go to the action.

Now

Now to Sep 2026 IMMEDIATE	Oct–Dec 2026 NEXT QUARTER	Jan–Mar 2027 NEXT PLANNING CYCLE

1 **Take a capital-committed position in Refusal-Layer flexibility tools before the next heat event.**

CSO / CFO

DECIDE

4 **Re-grade the capex thesis at the late-July Q2 prints, and watch merchant-solar capture-price deflation.**

CSO

MONITOR

◇ Late Jul: second-quarter results

3 **Reprice climate physical exposure through the credit channel, not the softening cat-cover market.**

CFO / CRO

PREPARE

◇ Q3: risk review

2 **Build a water-availability and interconnection-timing gate into siting and project-finance underwriting.**

COO / General Counsel

PREPARE

time to get ready ◇ key date DECIDE act now PREPARE get ready to act MONITOR watch it

Early warnings

Signals the portfolio surfaced before consensus formed, in the order they hardened.

LEAD TIME 2 WEEKS AHEAD · SIGNAL SCAN · DECISION INTELLIGENCE

The curtailable connection we flagged became a live emergency power.

INITIAL SIGNAL

We identified that US and British operators were re-pricing grid access around demand flexibility, conditioning fast interconnection on a large load's willingness to be curtailed.

NOW EVIDENT

- **FERC RM26-4**: a flexible-load fast lane that ties interconnection speed to curtailability, echoed by Texas's mandatory-curtailment law.
- **PJM, 30 June 2026**: two DOE 202(c) emergency orders authorising curtailment of data centres holding backup generation during the heatwave.
- **16 July verdict**: curtailment named as the Refusal Layer, with the authority rehearsed under heat stress though not yet executed.

CURTAILMENT ORDER **30 June 2026** STATUS **Authorised, not executed**

DECIDE

For the CSO and CFO, latent curtailment is now a priced condition on every large-load connection, not a tail scenario.

LEAD TIME 2 WEEKS AHEAD | SIGNAL SCAN | DECISION INTELLIGENCE

The water ceiling we flagged moved from footnote to permit veto.

INITIAL SIGNAL

We flagged that the binding water limit was not on-site cooling but the indirect water consumed at the power plants behind the electricity, a quantity an order of magnitude larger.

NOW EVIDENT

- **ITIF**: US data-centre indirect water reached roughly 211 billion gallons in 2023, more than ten times direct cooling.
- **Siting**: about two-thirds of post-2022 build sits in water-stressed basins, with 90.9% of the US Southwest in drought.
- **16 July verdict**: water reads as a permit veto ahead of the grid connection, a theme absent from the May register entirely.

INDIRECT WATER **211bn gallons (2023)** SOUTHWEST DROUGHT **90.9%**

PREPARE

For the COO and General Counsel, a site cleared on power and grid can still fail on a water permit landing after capital is committed.

LEAD TIME 6 WEEKS AHEAD | SIGNAL SCAN | CHANGE TRACKER | DECISION INTELLIGENCE

The factory ceiling we flagged, not the queue, set the delivery clock.

INITIAL SIGNAL

The early read placed the binding constraint downstream of permitting, in grid-hardware manufacturing: transformers, HV cables and switchgear carrying lead times up to four years on a concentrated electrical-steel base.

NOW EVIDENT

- **25 June tracker**: GE Vernova's gas-turbine backlog reached 100 GW, with slots booked toward 2030.
- **16 July ledger**: the supply-chain escalation graded confirmed, the transformer scramble holding.
- **Policy**: the US invoked the Defense Production Act, classing grid hardware as a national-defence input.

TRANSFORMER LEAD TIME **Up to 4 years** TURBINE SLOTS **Booked to 2030**

PREPARE

For the COO, the delivery clock now runs on manufacturing lead times that capital and permitting reform cannot compress.

LEAD TIME 5 WEEKS AHEAD | SIGNAL SCAN | DECISION INTELLIGENCE

The correlated stress we flagged, not the peak, decided the summer.

INITIAL SIGNAL

We flagged that a grid can hold an adequate reserve margin and still fail, because climate is correlating stress across generation, transmission and demand rather than testing the normal peak.

NOW EVIDENT

By the July verdict, adequate reserves were not enough: the grid cleared the 2 July peak only by leaning on emergency curtailment authority and demand response, while capacity markets stayed benchmarked to a static peak.

PEAK CLEARED **2 July 2026** CLEARED BY **Curtailment plus demand response**

PREPARE

For the CSO, reserve-margin adequacy alone understates reliability risk under correlated climate stress.

LEAD TIME 8 WEEKS AHEAD | SIGNAL SCAN | DECISION INTELLIGENCE

The load-follows-generation inversion we flagged set the colocation agenda.

INITIAL SIGNAL

The early read was that new load had begun choosing to sit next to generation, a structural inversion of the grid-strain story, anchored on FERC's December 2025 PJM colocation order.

NOW EVIDENT

Across the cycle the inversion hardened into policy: the colocation rulemaking advanced on the tracker ledger, and the behind-the-meter bypass became one branch of the July Refusal Layer as hyperscalers sited firm generation next to the load. On the firm-clean offtake window the May cycle flagged as closing, no grid-side reopening surfaced across the deliverables; the cycle's firm-clean response ran behind the meter rather than through new offtake on the network.

ORIGIN ORDER **FERC, December 2025** NOW **Colocation rulemaking advancing**

MONITOR

For the CSO, colocation and behind-the-meter siting is now a settled branch of the demand map, not a fringe case.

Assumption changes since the May scan

Cycle-over-cycle assumption changes, in decreasing order of strategic weight.

SHIFT 1 OF 5

"A sustained 15%+ hyperscaler capex cut would break the demand thesis"

PRIOR CYCLE HELD

The prior cycle held that the demand thesis had one clean disconfirming test: a sustained cut of 15% or more in hyperscaler capital guidance through Q3 would break it.

THIS CYCLE READS

The threshold broke upward, not down. The four hyperscalers each raised Q1 2026 guidance and are tracking roughly $725bn for the year, up about 77%; the buildout is confirmed, and the open question is returns, where a capex-to-revenue gap near $600bn now exceeds the 2001 telecom divergence.

FORWARD THESIS

Next cycle the fragile variable is returns, not demand: a Q2 or Q3 guidance break, not a demand miss, is the thing to watch.

SHIFT 2 OF 5

"The insurability frontier is hardening"

PRIOR CYCLE HELD

Prior thinking placed climate physical risk on a one-way hardening path, with a sixth consecutive $100bn-plus insured-loss year lifting the underwriting base.

THIS CYCLE READS

The frontier reversed on price while migrating venue. Mid-2026 catastrophe-reinsurance rates fell roughly 16 to 20% in an abundant-capacity buyers' market helped by a 2026 Atlantic hurricane outlook that turned below-normal, the steepest drop since the late 1990s, yet the stress surfaced in residual markets, with the California FAIR Plan at $724bn (up 230% since September 2022), and in household mortgage credit.

FORWARD THESIS

Next cycle the signal to watch is the credit channel, the premium-to-delinquency path, not the softening cat-cover price.

SHIFT 3 OF 5

"The binding constraint is the transmission and interconnection queue"

PRIOR CYCLE HELD

The prior cycle placed the binding constraint firmly in the interconnection queue, with roughly 1,400 GW of generation waiting on grid evacuation.

THIS CYCLE READS

The constraint would not hold still. It walked downstream across the cycle: from the queue to grid-hardware manufacturing, then to power-plant water, and finally to the grid's own authority to refuse load.

FORWARD THESIS

Next cycle the plan should be built against a moving constraint, not a fixed one, with the current binding edge re-identified each cycle.

SHIFT 4 OF 5

"Reliability equals adequacy, measured by the reserve margin"

PRIOR CYCLE HELD

Prior thinking equated reliability with adequacy: enough reserve margin held above the normal peak.

THIS CYCLE READS

Reliability is now carried by system strength, the inertia, fault current and reactive power procured explicitly, and by the right to refuse load. The 2 July peak held on curtailment authority and stability services, not on spare capacity.

FORWARD THESIS

Next cycle, reserve-margin adequacy alone will understate reliability risk; system strength and curtailment authority carry it.

SHIFT 5 OF 5

"Solar is the cheapest power and its build-out is unstoppable"

PRIOR CYCLE HELD

The prior consensus held solar as the cheapest power on an unstoppable build-out path, with cost the only variable that mattered.

THIS CYCLE READS

Solar's own success is collapsing its capture price. EU-27 day-ahead negative-price hours more than doubled year on year to 1,223 in Q1 2026, and close to half of April solar output in France and Germany cleared below zero, moving the binding constraint from generation cost to captured revenue.

FORWARD THESIS

Next cycle carries a new exposure the May thesis did not: the merchant-renewables revenue model, not the price of the panel.

Actions triggered

Each trigger names why it activates now, why it matters, the recommended actions, the primary decision owner, the decision window, and the conditions under which the recommendation should be reviewed.

DECIDE ACTION 1 SIGNAL SCAN DECISION INTELLIGENCE

Take a capital-committed position in Refusal-Layer flexibility tools before the next heat event.

TRIGGER CONDITION

This recommendation has been activated because PJM secured two DOE 202(c) emergency orders on 30 June 2026 authorising data-centre curtailment, because the curtailable-connection re-pricing has made flexibility the price of grid access, and because transformer and grid-hardware lead times of up to four years mean the supply the plan will need cannot be bought on demand.

PRIMARY DECISION OWNER **CSO / CFO**

DECISION WINDOW IMMEDIATE **Before the 2026 winter peak**

WHY THIS NOW MATTERS

Latent curtailment has moved from a tail scenario to a priced condition on every large load. The scarce assets that blunt it, flex capacity, backup, transformers and closed-loop cooling, are exactly the ones with the longest lead times.

WHO GAINS

The premium accrues to the suppliers of the scarce flexibility: grid-forming storage and inverter vendors, demand-response aggregators, backup-generation and closed-loop-cooling providers, and the transformer and grid-hardware makers holding near-term delivery slots.

RECOMMENDED ACTIONS

- **Immediate:** secure demand-response and flexible-capacity positions, and place transformer and grid-hardware orders against the four-year queue.
- **Immediate:** specify closed-loop cooling for new sites to cut both water and curtailment exposure.
- **Follow-on:** negotiate curtailable-connection terms that trade flexibility for interconnection speed.

REVIEW CONDITIONS

This recommendation should be reviewed if a second heat event forces actual curtailment execution, if FERC's co-location rulemaking resolves the terms of grid access, or if transformer lead times ease materially below three years.

PREPARE ACTION 2 SIGNAL SCAN DECISION INTELLIGENCE

Build a water-availability and interconnection-timing gate into siting and project-finance underwriting.

TRIGGER CONDITION

This recommendation has been activated because power-plant indirect water has surfaced as the binding siting limit, because 90.9% of the US Southwest sits in drought with two-thirds of post-2022 build in water-stressed basins, and because water rights are beginning to act as a permit veto ahead of the grid connection, a gate the current underwriting does not model.

PRIMARY DECISION OWNER COO / General Counsel

DECISION WINDOW NEXT PLANNING CYCLE 2027 capital plan

WHY THIS NOW MATTERS

A project cleared on power and grid access can still fail on a water permit, an exposure that lands after capital is committed. Pricing it at underwriting, not at permitting, is the difference between a repriced deal and a stranded one.

WHO GAINS

Value migrates to those who can clear the water gate: water-rights and hydrology advisers, closed-loop and air-cooling technology vendors, and siting and permitting consultancies, while owners of already-permitted, water-secure sites hold option value the market has not yet priced.

RECOMMENDED ACTIONS

- **Immediate:** add water-availability and indirect-water disclosure to site screening.
- **Next quarter:** set an interconnection-timing threshold below which projects do not pass finance.
- **Follow-on:** embed the gate in the 2027 capital plan as a standing underwriting condition.

REVIEW CONDITIONS

This recommendation should be reviewed if a permit-veto case blocks a peer's flagship site, if federal or state water-disclosure rules standardise the metric, or if drought conditions in the key basins ease durably.

PREPARE ACTION 3 SIGNAL SCAN DECISION INTELLIGENCE

Reprice climate physical exposure through the credit channel, not the softening cat-cover market.

TRIGGER CONDITION

This recommendation has been activated because catastrophe-reinsurance rates fell 16 to 20% mid-2026 in a buyers' market that masks the underlying risk, because the frontier migrated into residual markets such as the California FAIR Plan at $724bn, and because the Dallas Fed and Bank of England have traced a premium-to-delinquency path into household mortgage credit that the cat-cover price no longer signals.

PRIMARY DECISION OWNER CFO / CRO

DECISION WINDOW NEXT QUARTER Q3 2026 risk review

WHY THIS NOW MATTERS

Reading exposure off falling reinsurance prices would understate it precisely as the risk relocates into credit. The venue changed; the exposure did not close.

WHO GAINS

The repricing favours parametric and specialty underwriters, residual-market capacity, and the climate-risk analytics and rating providers that can price the credit channel the softening cat-cover market no longer signals.

RECOMMENDED ACTIONS

- **Immediate:** map physical-risk exposure onto residual-market and mortgage-credit indicators, not cat-cover pricing.
- **Next quarter:** fold the credit channel into the Q3 risk review and counterparty screens.
- **Follow-on:** stress-test asset and lending books against a residual-market and delinquency scenario.

REVIEW CONDITIONS

This recommendation should be reviewed if a residual market such as the FAIR Plan needs a public backstop, if reinsurance pricing reverses back upward, or if the mortgage-delinquency signal fails to broaden beyond the exposed basins.

MONITOR ACTION 4 DECISION INTELLIGENCE SIGNAL SCAN

Re-grade the capex thesis at the late-July Q2 prints, and watch merchant-solar capture-price deflation.

TRIGGER CONDITION

This recommendation has been activated because the Q2 2026 hyperscaler prints land after this synthesis publishes, because the capex-to-revenue gap near $600bn now exceeds the 2001 telecom divergence, and because solar's capture-price collapse (EU negative-price hours more than doubled year on year) is a new revenue-model exposure the May cycle did not carry.

PRIMARY DECISION OWNER CSO

DECISION WINDOW IMMEDIATE Late July 2026 Q2 prints

WHY THIS NOW MATTERS

The buildout is confirmed; the fragile variable is returns. A guidance break at the Q2 prints, or a deepening merchant-solar capture gap, is the trigger that would move this from watch to act.

WHO GAINS

The capture-price collapse rewards firming, not generation: grid-scale storage operators, flexibility and co-location developers, and firmed, storage-paired offtake, as unfirmed merchant renewables carry the revenue risk.

RECOMMENDED ACTIONS

- **Immediate:** set the named trigger, reinstate as Prepare if any hyperscaler cuts 2026 guidance by 15% or more at the Q2 prints.
- **Next quarter:** track EU day-ahead negative-price hours and solar capture rates as a merchant-revenue signal.
- **Follow-on:** pre-position a firmed, storage-paired stance to hedge merchant-renewables exposure.

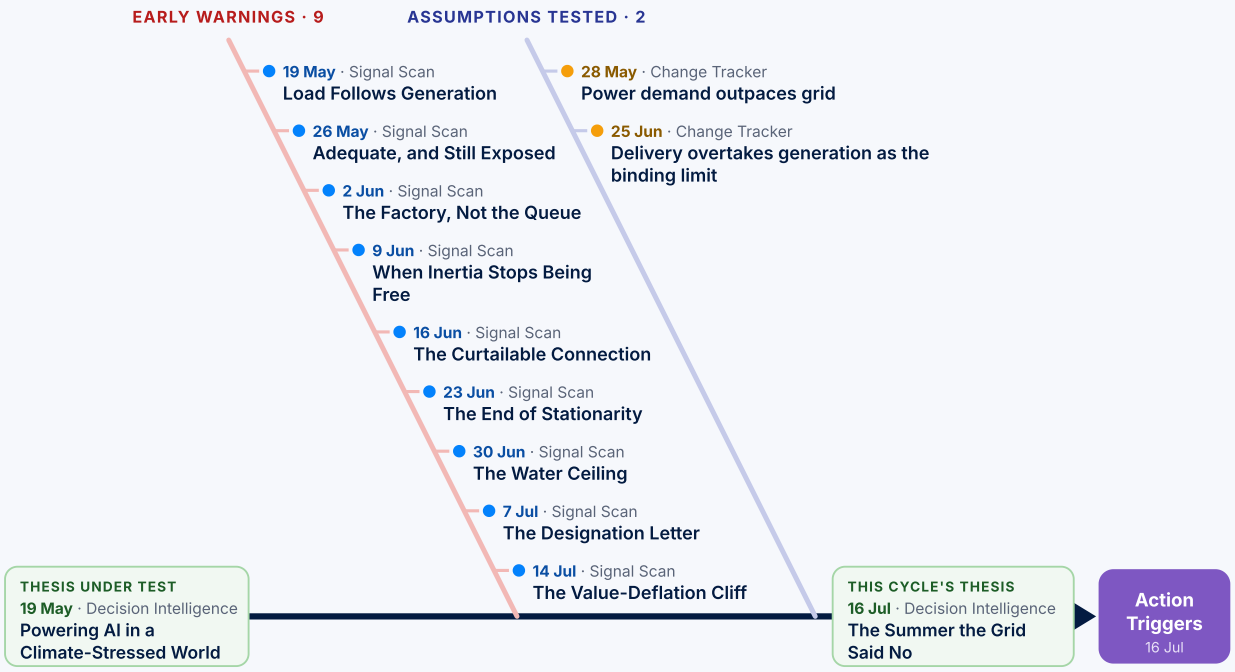
REVIEW CONDITIONS

This recommendation should be reviewed if Q2 or Q3 guidance breaks the 15% threshold (escalate to Prepare), if the capex-to-revenue gap narrows toward historical norms, or if negative-price hours stabilise on current European storage deployment.

IF YOU ACT ON ONE THING

Commit capital to flexibility, backup power and cooling before next winter's peak, because grid access now depends on being able to cut load on demand.

DRAWING ON



Signal Scan · Change Tracker · Decision Intelligence

Each bone groups the deliverables by what they added, with the oldest at the outer end. Hover over any deliverable for its full title.

HOW THIS WAS MADE

The Decision-Grade Intelligence Method

Shaping Tomorrow's six-stage process for converting external change into decision-ready intelligence.

Scan · Filter · Stress-test · Judge · Trigger · Actionable Intelligence

See full methodology → · How we verify → · Track record →

This briefing is not another report. It is the synthesis produced after continuously monitoring signals, tracking change, testing assumptions and updating scenarios

throughout the cycle on **Energy, Infrastructure & Climate Resilience**. Its purpose is to identify the points at which accumulated evidence has become decision-changing: the early warnings the work caught before consensus formed, the assumptions that have shifted between cycles, and the actions now decision-shaped enough to act on. Continuous intelligence on this topic is available as a Single-Topic or Multi-Topic programme.

Discuss a Tailored Programme

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